Amazon ML Challenge 2025

Team Name: Team@Alpha

Team Members:

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1 Executive Summary.

A Streamlit app for product price prediction uses both catalog text and product image data for improved accuracy.**2. Methodology Overview**

2.1 Problem Analysis

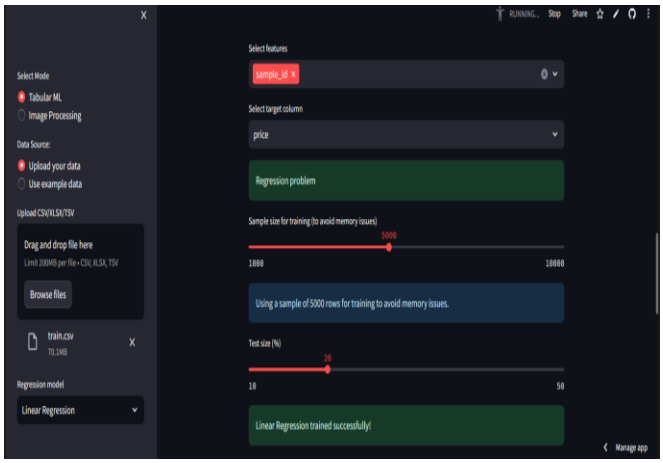
Predicting prices from mixed tabular and image data requires extracting relevant features from both sources and combining them effectively.

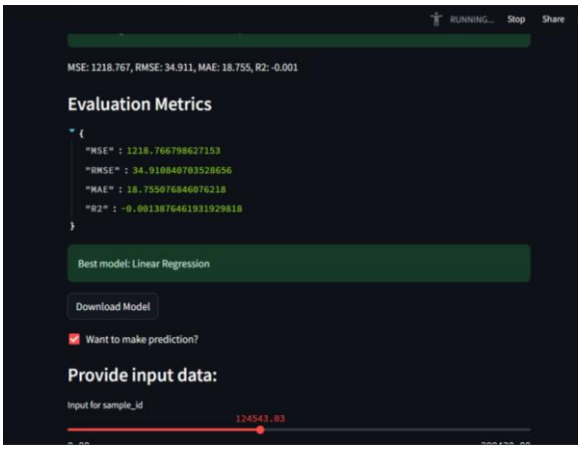
2.2 Solution Strategy

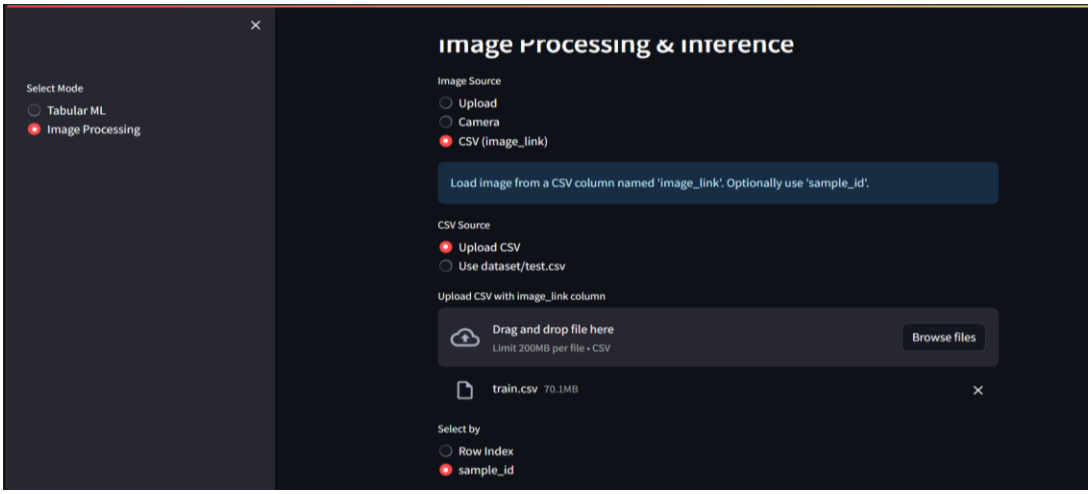
- Hybrid model: Uses TFIDF + Ridge for text and Vision Transformer + Linear Regression for images.
- Ensemble: Averages predictions from both if available; otherwise, uses one.

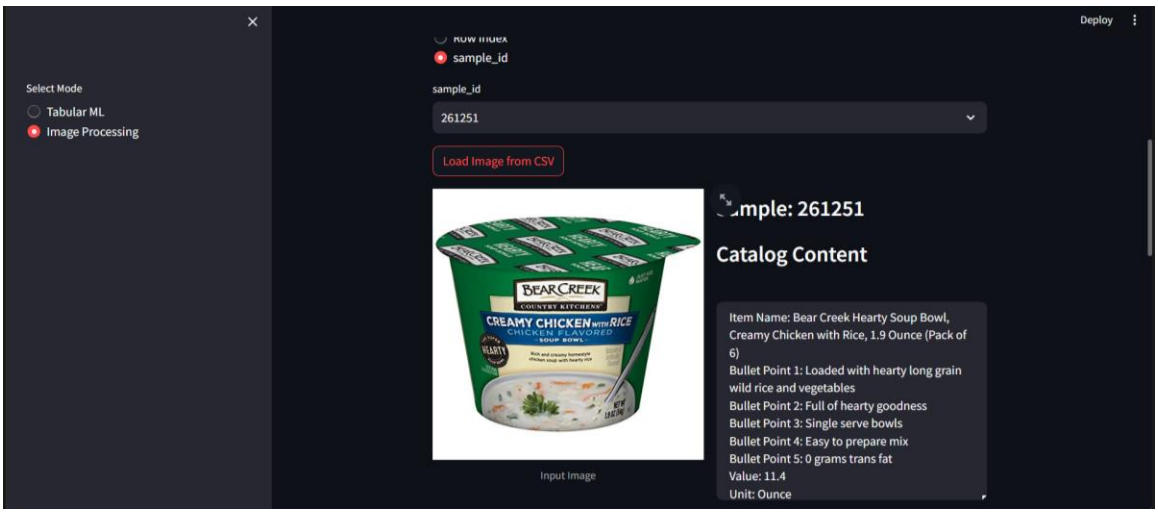
3. Model Architecture

- **Text:** TFIDF → Ridge Regression
- **Image:** Preprocess → ViT Embedding → Linear Regression
- **Combined:** Average of both models' outputs









4. Model Performance

1. Regression: SMAPE, MAE, RMSE, R^2 (reported in-app results).
2. Classification: Accuracy, precision, recall, F1, AUROC.
3. Best model highlighted in UI automatically.

5. Conclusion

Code drive link:

- https://drive.google.com/drive/folders/1WeVwPx10s10ATixfn-MSajSqI6Nyxfil?usp=sharing
- Additional Results:
    - Interactive plots, downloadable models/images, error messages for data/memory issues